

COURSE TITLE : CAD Lab II
COURSE CODE : 5189
COURSE CATEGORY : A
PERIODS/WEEK : 3
PERIODS/SEMESTER : 39
CREDITS : 2

TIME SCHEDULE

MODULE	TOPIC	PERIODS
I	Introduction to 3D modelling	4
II	Modifying commands and mesh modelling	20
III	Editing commands	9
IV	Rendering and Lighting	6
TOTAL		39

Course Objective :

Sl.	Sub	Student will be able to
1	1	Develop 3D views from 2D drawings
	2	Apply materials and shading on 3D views
	3	Export and import data

Upon completion of the course the student should be able to:

MODULE –I— Introduction to 3D modelling

1.1.0. Familiarize with 3D modeling

- 1.1.1. Know types of 3D Models
- 1.1.2. Know 3D Coordinate Systems

- 1.1.3. Know Dynamic Viewing of 3D Objects
- 1.1.4. Create Predefined Solid Primitives

MODULE—II—Modifying commands and mesh modelling

2.1.0. Familiarize modifying commands

- 2.1.1. Use modifying commands like extrude, sweep, revolve, loft etc.. .

2.2.0. Familiarize mesh modeling

MODULE—III-- Editing commands

3.1.0. Understand editing commands

- 3.1.1. Use editing commands like union , subtraction, intersection, slice etc..

3.2.0. Create 3D models of buildings

MODULE—IV—Rendering, mapping and Lighting

4.1.0. Familiarize with rendering, mapping and lighting

CONTENT DETAILS

MODULE –I— Introduction to 3D modelling

Types of 3D Models-3D Coordinate Systems- tool palettes –palettes for solid modelling, render, visual styles, view etc. Models -Creating Predefined Solid Primitives

MODULE—II-- Modifying commands and mesh modelling

Complex 3D Geometry-Modifying commands -Extrude, sweep, revolve, loft, shell etc- Mesh Modeling - Mesh Primitives

MODULE—III-- Editing commands

Creating Complex Solid Models by Applying Boolean Operations –union, intersection, subtraction etc.. Creating Surface Meshes-Editing Faces of Solids--Fillet and Chamfers on Solids-modeling of building

MODULE—IV—Rendering, mapping and Lighting

Concept of Rendering - Elementary rendering -mapping-types of mapping-Selecting and Attaching Materials -Adding Lights to the Design

REFERENCE BOOKS

- 1. AutoCAD - Dayanithi
- 2. AutoCAD with Application - Sham Tickoo